Daily Logic Drill #4

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

*New toolkit today: Disproving Universal Claims · Minimal Counterexamples · Common Counterexample Sources (*0, 1, −1*, non-integers).*

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤ 15 s each)

Kill a universal claim with a single well-chosen example. Pure recognition.

1. Give a counterexample to: "Every prime number is odd."

2. Give a counterexample to: "$n^2 \geq n$ for every real number n."

3. Give a counterexample to: "If n is prime, then $n + 2$ is prime."

4. Give a counterexample to: "The sum of two primes is always even."

5. Give a counterexample to: "Every multiple of 4 is a multiple of 8."

6. Give a counterexample to: "$\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ for all $a, b \geq 0$."

7. Give a counterexample to: "$(x + y)^2 = x^2 + y^2$ for all real x, y."

8. Give a counterexample to: "$n! + 1$ is always prime."

9. Give a counterexample to: "If a number is not prime, it must be even."

10. Give a counterexample to: "All odd numbers greater than 1 are prime."

Section B Manipulation Drills (≤ 50 s each)

Hunt systematically, and notice where 0, 1, −1, *and non-integers do the most damage.*

1. Give a counterexample to: "$n^2 + n + 41$ is prime for every positive integer n."

2. Give a counterexample to: "If a and b are both irrational, then $a + b$ is irrational."

3. Give a counterexample to: "If a and b are both irrational, then ab is irrational."

4. Give a counterexample to: "Every quadratic with integer coefficients that has an integer root must have two integer roots."

5. Give a counterexample (using $n = 0$) to: "For every integer n, $n/n = 1$."

6. Give a counterexample (using $n = 1$) to: "Every positive integer has at least one prime factor."

7. Give a counterexample (using $n = -1$) to: "For all integers n, n^n is positive."

8. Give a counterexample (using a non-integer) to: "If x^2 is an integer, then x is an integer."

9. Give a counterexample to: "If p and q are distinct primes, then $p + q$ is composite."

10. Give a counterexample to: "For every integer $n \geq 1$, $2^n - 1$ is prime."

Section C Substitution & Structure (≤90 s each)

Find the smallest counterexample, not just any counterexample — and spot a structural trap.

1. Find the smallest positive integer n for which $n^2 + n + 41$ is *not* prime.

2. Give a counterexample to: "If $a \equiv b \pmod{n}$, then $a^2 \equiv b^2 \pmod{n^2}$."

3. Which of the following values of n is a counterexample to "$n^2 - n + 11$ is prime for every positive integer n"?

A) $n = 1$

B) $n = 5$

C) $n = 11$

D) $n = 10$

4. Give a counterexample to: "If p is prime, then $2^p - 1$ is prime."

5. Give a counterexample to: "If $a^2 \equiv b^2 \pmod{n}$, then $a \equiv b \pmod{n}$."

6. Find the smallest positive integer n for which $3^n + 2$ is *not* prime.

7. Give a counterexample to: "If x and y are both non-integers, then $x + y$ is a non-integer."

8. Give a counterexample to: "For all integers a, b, c: if $a \mid bc$, then $a \mid b$ or $a \mid c$."

Section D Challenge (TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Consider the claims: I. "$n^2 + n + 41$ is prime for every positive integer n." II. "$2^n - 1$ is prime whenever n is prime." III. "$n! + 1$ is prime for every positive integer n." Which of I, II, III are true?

A) I only

B) II only

C) III only

D) I and II only

E) I and III only

F) II and III only

G) I, II and III

H) None of them

2. Find the smallest positive integer n for which $n^2 + n + 1$ is *not* prime.

3. Give a counterexample to: "The sum of any three consecutive prime numbers is always even."

4. Give a counterexample to: "If n is a positive integer such that $n^2 + 1$ is prime, then n must be even."

5. Statement T (you may take this as given/true): "If $2^n - 1$ is prime, then n is prime." (a) Explain why B10 and C4's counterexamples do *not* contradict T. (b) State T's converse, and determine whether it is true. (c) Is "$2^n - 1$ is prime" necessary and sufficient for "n is prime"?

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths